Centers for Disease Control and Prevention

Vaccine Safety

Understanding Adverse Events and Side Effects

Safety information on recommended vaccines

CDC provides information about common and rare adverse events that may occur after vaccination.

Vaccine Information Statements (VISs)

Safety information by Vaccine

Any health problem that happens after vaccination is considered an adverse event following immunization. An adverse event can be a true adverse reaction, also known as a side effect, that is related to the vaccine, or a coincidental event that happened following vaccination.

CDC's Immunization Safety Office, along with FDA and other federal government partners, conducts research on adverse events that occur after vaccination.

Types of adverse events that may occur after vaccination

Adverse Reactions (side effects)

Where side effects occur in the body

- **Local reactions** occur where the shot was given. They are usually mild and occur within a few hours of the shot. Injections site pain, redness or swelling are most commonly reported local adverse reactions.
- **Systemic reactions** include fever, headache, body aches, fatigue, etc. These reactions are usually mild and can last for several days.

Source: Chapter 2 General Recommendations on Immunization. [Epidemiology and Prevention of Vaccine Preventable Diseases. 2015].

Adverse reactions, also known as **side effects,** are considered to be caused by a vaccine. Usually, vaccine side effects are identified during clinical trials. The intensity of these reactions may range from mild to moderate to severe. They often resolve on their own, and may or may not require medical intervention. Depending on severity, an adverse reaction may also be considered a serious adverse event.

Unrelated health problems following vaccination

People can experience health problems following vaccination that would have occurred even if the person was not vaccinated. These health problems are not related to the vaccine.

Health problems with an unknown cause

People can experience health problems following vaccination, but the cause of these events are unknown. There is not enough evidence to say whether it was caused by the vaccine.

Reporting adverse events to VAERS

The Vaccine Adverse Event Reporting System (VAERS) is the nation's early warning system that monitors the safety of vaccines after they are authorized or licensed for use by the U.S. Food and Drug Administration (FDA). Reports to VAERS of adverse events are classified as non-serious or serious:

Serious adverse event report — These reports meet the definition of "serious" specified by the Code of Federal Regulations because one of the following is reported: death, life-threatening illness, hospitalization or prolongation of hospitalization, permanent disability, congenital anomaly, or birth defect.

Non-serious adverse event report — These reports do not meet the regulatory definition of a serious adverse event report.

Learn more about reporting adverse events to VAERS.

Related Scientific Articles

Shimabukuro TT, Nguyen M, Martin D, DeStefano F. Safety Monitoring in the Vaccine Adverse Event Reporting System (VAERS). *Vaccine.* 2015 Aug 26;33(36): 4398-405. Epub 2015 Jul 22.

Council for International Organizations of Medical Sciences (CIOMS)/ World Health Organization (WHO) Working Group on Vaccine Pharmacovigilance. Definition and Application of Terms for Vaccine Pharmacovigilance. [PDF – 198 Pages] *WHO Press*. 2012.

Shimabukuro TT, Cole M, Su JR. Reports of Anaphylaxis After Receipt of mRNA COVID-19 Vaccines in the US—December 14, 2020-January 18-2021. *JAMA*. 2021 Feb 12. doi:10.1001/jama.2021.1967 Epub ahead of print.

More Information

Reporting Adverse Events

What adverse events to report to VAERS, how to submit reports, and why.

Vaccine Safety Research

The latest safety research on U.S.-licensed vaccines.

The Journey of Your Child's Vaccine

A video on the life cycle of a vaccine, from development through the safety monitoring after it has been added to the immunization schedule.

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